

# Clause-Activation Signatures (CAS) for Attack-Family Attribution in IoMT

Rakesh Reddy Yakkati, Per-Arne Andersen, Lei Jiao, Ole-Christoffer Granmo, and Rishad Shafik

**Abstract**—When an intrusion detector flags traffic in the Internet of Medical Things (IoMT), a label alone is not enough — the analyst must also know *which* attack family is at play and *why*. This paper introduces Clause-Activation Signatures (CAS), a method that turns the clauses of a trained weighted Tsetlin Machine (TM) into a compact, auditable signature for each attack family, so that attribution becomes signature matching rather than an opaque vote. Each signature is extracted with complementary-pairs stability selection (CPSS) over clause activations, split by coefficient sign into inculpatory and exculpatory evidence, and certified by a Shah–Samworth bound on spurious inclusions. Evaluated on CICIoMT2024 (WiFi/MQTT, family level) across three independently seeded TMs, CAS attribution improves over the raw weighted-vote TM it is extracted from: macro-F1 rises from  $0.692 \pm 0.004$  to  $0.765 \pm 0.029$  at the verified operating point, using only 28–53 sign-positive clauses per family out of a 2,400-clause pool. Sorting each class’s clauses by weight and discarding the weakest 84% before extraction costs the base detector nothing, lifts CAS to  $0.779 \pm 0.025$ , and removes its sensitivity to the stability threshold entirely. The six signatures provably form a maximally 5-disjunct code on every seed, so any mixture of up to five families decodes exactly; under 5% bit-flip noise a thresholded decoder still recovers 99.8%–100% of mixtures. A binary normal-versus-attack reduction lifts AUROC from 0.819 to 0.940. Two honest negative results are reported: open-set novelty detection is weak under correct conformal false-flag control, and signature identity does not survive retraining under a different seed. CAS is thus a certified attribution and interpretability layer that denoises a mediocre base detector, not as a replacement for one.

**Index Terms**—Tsetlin Machine, interpretable machine learning, intrusion detection, IoMT, stability selection, disjunct codes, attack attribution, open-set recognition

## I. INTRODUCTION AND RELATED WORK

Intrusion detection for the Internet of Medical Things (IoMT) must satisfy two requirements simultaneously: the alarm must be correct, and it must identify the responsible attack family in a form a forensic analyst can audit, since the appropriate response differs fundamentally across families

This work was supported by the *SecureIoTM—Ultra-low-energy IoT Intrusion Detection Systems using Logic-based Tsetlin Machines* project (Grant Number 342167), funded by the Research Council of Norway under the TEKNOLOGIKONVER programme. The authors acknowledge the support from the Centre of Artificial Intelligence Research at the University of Agder, Sørlandet Hospital Trust, and Newcastle University for their collaboration in this interdisciplinary research initiative.

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and patient safety can depend on the distinction. Existing IDS explainability does not meet the second requirement. Post-hoc explainers such as SHAP, LIME, and Anchors [14] account for one prediction at a time and provide no stability guarantee: retraining the model or perturbing the explainer changes the explanation. A per-family *signature* — a small rule set that is interpretable by design, vetted for stability, and attached to a family rather than to a single flow — remains absent from the literature.

A candidate substrate for such signatures is the Tsetlin Machine (TM) [1], whose unit of computation is a conjunctive clause over Boolean literals learned through bandit-driven Type I/II feedback; clauses remain human-readable after training, and the weighted TM [2] preserves this property while learning an integer vote weight per clause. TM-based intrusion detection is established: Abeyrathna et al. extracted interpretable TM rules on KDD’99/NSL-KDD [6], and Jaiswal et al. trained TMs on CICIoMT2024, reporting 90.7% six-class accuracy [3] and 97.8% macro-F1 on MedSec-25 [7]. These works, however, explain individual predictions through whichever clauses happened to fire; they do not aggregate activations into stable per-family signatures, and they provide no error control on the explanation.

This paper closes that gap with Clause-Activation Signatures (CAS) for attack-family attribution. A trained weighted TM maps every flow to a vector of clause activations. For each family  $\ell$ , CAS selects the clauses whose activations are stably predictive of  $\ell$  using complementary-pairs stability selection (CPSS) [4] over the clause-activation space, and splits the selection by coefficient sign into inculpatory and exculpatory evidence. Because CPSS is governed by the Shah–Samworth bound, each signature carries a certificate on the expected number of spurious inclusions. Attribution then reduces to signature matching: a flow is attributed to the family whose signature it matches best, and the matched clauses are the explanation returned to the analyst.

The construction rests on two bodies of theory. The first is stability selection [4], [8], applied here to clause activations rather than raw features; to our knowledge this usage, and clause-level forensic security signatures generally, are new. The second is  $t$ -disjunct matrices from combinatorial group testing [15]: if no column of a binary incidence matrix is covered by the union of any  $t$  other columns, any mixture of at most  $t$  columns decodes exactly.

The evaluation requires a dataset that survives standard IDS validity criteria [9]–[11]. CICIoMT2024 [12] is the only peer-reviewed, widely adopted IoMT-specific benchmark spanning

multiple protocols, and its schema contains no IP or port fields, removing the most common leakage channel by design.

The contributions are: (1) a correctness-audited, seed-reproducible CAS implementation on TMU [5] clause activations, including a sign-filtering fix without which attribution falls below the base detector (§II-F, §IV-A); (2) a three-seed verified evaluation showing CAS outperforms its base TM above a stability threshold we identify (§IV-B); (3) signed, human-readable evidence lists per family with an explicit error-control certificate (§IV-E); (4) an exact disjunct-code certificate, maximal on all seeds, where the decoder rather than the code is the noise bottleneck (§IV-H); (5) a binary normal-versus-attack reduction where the attack signature markedly improves detection ranking (§IV-D); and (6) two negative results: weak open-set novelty power under correct conformal false-flag control (§IV-G), and near-chance cross-seed signature identity (§IV-I).

## II. METHOD: CLAUSE-ACTIVATION SIGNATURES

### A. Overview

Fig. 1 shows the core stages involved in detail for the proposed method. Initially, the raw flows are binarized using existing standard binarizer in TMU library. Then, a weighted TM is trained on them. Then the final output of the TM is its learned rules (clauses that fire on it). The purpose of CPSS is to extract a stable clauses per class. In this process, two stages accompany the clauses: a statistical error-control bound and a combinatorial identifiability guarantee.

### B. Dataset Description

We use CICIoMT2024 [12], WiFi/MQTT branch, with family-level labels. The dataset was collected on a testbed of 40 healthcare devices (25 real, 15 simulated) communicating over Wi-Fi, MQTT, and BLE. It contains 18 attack types grouped into five families, and each flow is described by 22 numeric CIC-extracted flow features. The six families are Benign, distributed denial-of-service (DDoS), denial-of-service (DoS), attacks on the Message Queuing Telemetry Transport protocol (MQTT), reconnaissance (Recon), and Spoofing.

### C. Booleanization and Base Detector

Each numeric flow feature is discretized into 10 quantile bins. We use scikit-learn’s KBinsDiscretizer with bin edges fit on the training partition only. The bins are encoded as cumulative thermometer bits of the form “feature  $\geq$  bin  $k$ ”. TMU handles literal negation internally.

### D. WTM

We train a weighted multiclass TM on the six classes with 400 clauses per class,  $T = 320$ ,  $s = 8.0$ , and 25 epochs. Since, the dataset has six attack-family classes. Passing flows through the trained TM, with 400 clauses learned per class, yields the clause-activation matrix  $Z \in \{0, 1\}^{n \times 2400}$ . Here  $Z[x, j] = 1$  if and only if clause  $j$  fires on input  $x$ .

### E. Clause-Pool Pruning

Not every clause in the pool is worth carrying into signature extraction, and which ones to drop can be posed exactly. Write  $w_{c,j}$  for the learned weight of clause  $j$  of class  $c$  and  $z_{c,j}(x) \in \{0, 1\}$  for its activation, so the class sum is  $f_c(x) = \sum_j w_{c,j} z_{c,j}(x)$  and  $\hat{y}(x) = \arg \max_c f_c(x)$ . A keep-mask  $m \in \{0, 1\}^{C \times n}$  with deletion sets  $D_c = \{j : m_{c,j} = 0\}$  gives  $f_c(x; m) = \sum_j m_{c,j} w_{c,j} z_{c,j}(x)$ , and we seek the smallest pool that preserves accuracy:

$$\min_{m \in \{0, 1\}^{C \times n}} \|m\|_0 \quad \text{s.t.} \quad F_1(\hat{y}(\cdot; m)) \geq F_1(\hat{y}(\cdot; \mathbf{1})) - \varepsilon. \quad (1)$$

This is a search over  $2^{Cn}$  masks ( $2^{2400}$  here) whose constraint is non-decomposable, macro-F1 sitting behind an  $\arg \max$ . We therefore optimise the perturbation it induces,  $R(D_c) := \sum_{j \in D_c} |w_{c,j}|$ , which controls (1) through the decision margin.

**Proposition 1** (Certificate). *Let  $R = \max_c R(D_c)$  and let  $\gamma(x) = f_{\hat{y}(x)}(x) - \max_{c \neq \hat{y}(x)} f_c(x)$  be the full-pool margin. If  $\gamma(x) > 2R$  then  $\hat{y}(x; m) = \hat{y}(x)$ .*

*Proof.* Since  $z_{c,j}(x) \in \{0, 1\}$ ,  $|f_c(x) - f_c(x; m)| = |\sum_{j \in D_c} w_{c,j} z_{c,j}(x)| \leq R(D_c) \leq R$  for every  $c$ . Writing  $a = \hat{y}(x)$ , for any  $c \neq a$ ,  $f_a(x; m) - f_c(x; m) \geq (f_a(x) - R) - (f_c(x) + R) \geq \gamma(x) - 2R > 0$ , so  $a$  remains the maximiser.  $\square$

**Proposition 2** (Exact greedy). *For a budget  $|D_c| = k$ ,  $\min_{D_c} R(D_c)$  is attained by the  $k$  clauses of smallest  $|w_{c,j}|$ .*

*Proof.*  $R(D_c)$  is a sum of  $k$  non-negative terms selected without replacement. If  $D_c$  contains  $j$  while excluding  $j'$  with  $|w_{c,j'}| < |w_{c,j}|$ , swapping them strictly decreases  $R$ ; no such swap exists precisely when  $D_c$  holds the  $k$  smallest magnitudes.  $\square$

Magnitude ranking is therefore optimal for the surrogate rather than a heuristic, and by Proposition 1 it carries a per-input guarantee, at cost  $O(Cn \log n)$  against the  $2^{Cn}$  of (1). Sweeping  $k$  in steps of five and taking the largest  $k$  that meets the constraint gives the operating budget used in §IV-F; CAS is then extracted from the surviving pool exactly as below, with nothing else changed.

### F. CPSS Signature Extraction and Sign Filtering

CAS works entirely in clause activation space, where a flow is represented by the rules that match it. We build one one-versus-rest problem per family  $\ell$ . CPSS then repeats the following procedure  $B$  times. It splits a validation half of the test set into two complementary halves. On each half it fits an  $\ell_1$ -regularized logistic model. The features of this model are the clause activations, i.e., the columns of  $Z$ . The target is the one-versus-rest label of family  $\ell$ . This yields  $2B$  half-fits per family.

Two knobs control this stage. The first is  $\kappa$ . It is the inverse-regularization strength of the logistic model inside each half-fit. The second is  $\pi_{\text{thr}}$ . It is the stability threshold applied after all fits are done. The knob  $\kappa$  sets how sparse one single fit

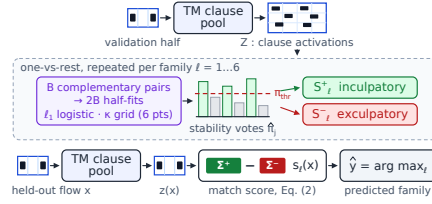


Fig. 1. CPSS signature extraction and sign filtering. CPSS votes on clauses across  $2B$  complementary half-fits (a  $\kappa$  grid per fit), thresholds the votes at  $\pi_{\text{thr}}$ , and splits each signature by coefficient sign into inculpatory  $S_{\ell}^+$  and exculpatory  $S_{\ell}^-$  evidence for scoring.

is. A small  $\kappa$  forces most coefficients to exactly zero. Each coefficient belongs to one clause. Only a few clauses then survive a single half-fit. We do not fix one  $\kappa$ . We sweep it over a geometric grid of 6 points from 0.001 to 0.1. A clause counts as selected by a half-fit if any grid point selects it. This union-over-grid rule follows [4].

Each half-fit casts one vote per clause. The stability score  $\hat{\pi}_j$  is the fraction of the  $2B$  half-fits that select clause  $j$ . The signature keeps the clauses that win enough votes:  $S_{\ell} = \{j : \hat{\pi}_j \geq \pi_{\text{thr}}\}$ . So  $\pi_{\text{thr}}$  sets how consistent a clause must be across votes. The two knobs are not independent.  $\kappa$  sets which clauses can collect votes at all.  $\pi_{\text{thr}}$  then cuts the resulting vote distribution. A denser  $\kappa$  grid lets more clauses survive each fit. More clauses then pass any fixed  $\pi_{\text{thr}}$ . So the signature size at a given threshold depends on the grid. A low  $\pi_{\text{thr}}$  threshold admits clauses that barely pass half of the fits. A higher threshold keeps only clauses that nearly every fit selects. We restrict  $\kappa$  to the range that keeps each half-fit sparse on this 2,400-clause pool. A wider grid would inflate  $q_{\Lambda}$  and make the certificate vacuous.

One further step is needed. CPSS selects on  $|\text{coefficient}| > 0$ . This is correct for the error-control bound. It is wrong for attribution. A stably negative coefficient is evidence *against*  $\ell$  and not for it. We therefore track the mean signed coefficient  $\text{coef}_j$  of each clause across fits. We split the signature into  $S_{\ell}^+ = \{j \in S_{\ell} : \text{coef}_j > 0\}$  and  $S_{\ell}^-$ . The positive part is inculpatory evidence. The negative part is exculpatory evidence. The certificate still applies to the full sign-agnostic  $S_{\ell}$ . We state this scoping gap openly. §IV-A shows that this split is not cosmetic.

The attribution score is

$$s_{\ell}(x) = \frac{\sum_{j \in S_{\ell}^+} \hat{\pi}_j z_j(x) - \sum_{j \in S_{\ell}^-} \hat{\pi}_j z_j(x)}{\sum_{j \in S_{\ell}^+} \hat{\pi}_j}, \quad (2)$$

and the prediction is  $\hat{y} = \arg \max_{\ell} s_{\ell}(x)$ . Matches to inculpatory clauses add weight. Matches to exculpatory clauses subtract weight. Normalization by the inculpatory weight is required. Without it scores scale with signature size instead of match quality.

### G. Error-Control Certificate

With  $q_{\Lambda}$  estimated as  $\text{mean}(\hat{\pi}) \cdot p$ , the Shah–Samworth bound gives  $E[V] \leq q_{\Lambda}^2 / ((2\pi_{\text{thr}} - 1)p)$  with  $p = 2400$ . We report these bounds as loose wherever they are loose (§IV-E).

They certify that most of each signature is likely genuine. They do not certify that it is near-noise-free.

## III. EXPERIMENTAL SETUP

We apply stratified caps for tractability. Each family contributes at most 40,000 training and 15,000 test rows. Benign (19,291 rows) and Spoofing (16,047 rows) are used in full. This yields 195,338 training and 80,868 test rows. The raw distribution is heavily imbalanced, with DDoS and DoS together accounting for about 95.7% of flows [12]. We therefore report macro-F1 and per-class metrics throughout.

Signatures are constructed with CPSS on a random half of the test set and evaluated on the disjoint remaining half, so no leakage occurs between signature fitting and evaluation. Multi-seed results use three independently trained TMs with seeds 42, 7, and 123 and fully seeded CPSS.

## IV. RESULTS

We proceed from construction checks (§IV-A) to closed-set family attribution (§IV-B), its sensitivity to the CPSS knobs (§IV-C) and its binary reduction (§IV-D), the signatures themselves (§IV-E) and what happens when the clause pool is pruned first (§IV-F), open-set behavior (§IV-G), mixture identifiability (§IV-H), and cross-seed stability (§IV-I).

### A. Validation of the Construction

Three checks validate the construction. First, the Rate-clipping fix (§II-C) is essential: without it, macro-F1 collapses from 0.72 to 0.54. Second, the sign split (§II-F) is not cosmetic: a naive sign-agnostic scorer reaches only 0.642–0.666 macro-F1 — *below* the 0.686 TM-argmax baseline — while the signed scorer reaches 0.719–0.776, comfortably above it. Third,  $\pi_{\text{thr}} = 0.6$  is unstable: reruns score as low as 0.638–0.643 with cross-seed standard deviation 0.028–0.032, roughly twice the 0.015–0.022 at  $\pi_{\text{thr}} \geq 0.7$ ; at 0.8 the selected set is stably separated. We therefore report  $\pi_{\text{thr}} = 0.8$  as the verified operating point throughout.

### B. Task A: Closed-Set Family Attribution

Table I gives the verified three-seed headline results: at the verified threshold CAS beats its base detector on every seed, with smaller signatures than at lower thresholds; raising  $B=15$  to  $B=25$  and fusing with the raw TM vote add nothing further.

The gains come from where the base detector is weakest: in the recorded primary run (seed 42,  $\pi_{\text{thr}}=0.6$ ,  $B=15$ ), Benign recall rises from 0.30 to 0.64 and Spoofing precision

TABLE I  
TASK A: VERIFIED THREE-SEED MACRO-F1 ON CICIOMT2024.

| Method                              | $B=15$                              | $B=25$            |
|-------------------------------------|-------------------------------------|-------------------|
| Weighted-TM argmax (base)           | $0.692 \pm 0.004$                   | (same models)     |
| CAS $\pi_{\text{thr}}=0.6$          | $0.688 \pm 0.032$                   | $0.678 \pm 0.028$ |
| CAS $\pi_{\text{thr}}=0.7$          | $0.743 \pm 0.015$                   | $0.737 \pm 0.022$ |
| CAS $\pi_{\text{thr}}=0.8$          | <b><math>0.765 \pm 0.029</math></b> | $0.750 \pm 0.034$ |
| CAS $\pi_{\text{thr}}=0.9$          | $0.762 \pm 0.038$                   | $0.761 \pm 0.038$ |
| Fusion (TM + CAS, $\lambda$ on val) | $0.764 \pm 0.010$                   | $0.768 \pm 0.013$ |

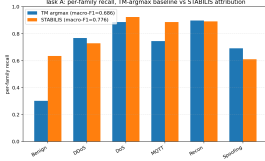


Fig. 2. Per-family recall, base weighted-TM argmax versus CAS attribution; the gains concentrate on Benign and Spoofing.

from 0.41 to 0.55 (Fig. 2). DDoS and DoS, by contrast, stay mutually confused under both methods, with near-identical F1 — a documented dataset property [13], not a signature-quality artifact.

### C. Sensitivity to $\pi_{\text{thr}}$ , $B$ , and $\kappa$

Fig. 3 sweeps the two CPSS knobs and the number of complementary pairs, on the same three seeds and the same evaluation half as Table I. Both metrics tell the same story about  $\pi_{\text{thr}}$ : at 0.5–0.6 CAS sits at or below the base detector (macro-F1 0.708/0.688 vs. 0.692; accuracy 0.731/0.712 vs. 0.726), from 0.7 it clears the base on both, and it peaks at  $\pi_{\text{thr}}=0.8$  (0.765 macro-F1, 0.783 accuracy) — the operating point adopted throughout. The number of complementary pairs matters much less: macro-F1 rises from 0.738 at  $B=5$  to 0.765 at  $B=15$  and is flat thereafter (0.750 at  $B=20$  and  $B=25$ ), so  $B=15$  already saturates the stability estimate and every setting stays above the base detector.

The regularization strength is the least sensitive knob (Table II): across two orders of magnitude,  $\kappa \in [10^{-3}, 4 \times 10^{-2}]$ , the three-seed mean varies only between 0.725 and 0.764, and it is only at  $\kappa=0.1$  — where single half-fits stop being sparse — that CAS degrades toward the base detector (0.674–0.722). This is what justifies the union-over-grid rule of §II-F: at the verified threshold the union (0.765) beats every individual grid point (best single  $\kappa$ : 0.752), so pooling the grid recovers clauses no single sparsity level selects on its own.

### D. Binary Reduction: The Attack Signature

Attribution helps only after a flow is flagged malicious, so we next ask what CAS offers the simpler normal-versus-attack problem: collapsing the five attack families into one ATTACK class and training a dedicated two-class TM yields a binary CAS. Table III and Fig. 5 tell a clear story. The base TM over-predicts attack: it catches almost everything (recall 0.988) but cries wolf often (precision 0.847). The attack signature

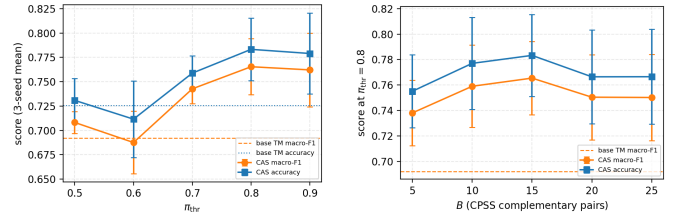


Fig. 3. Sensitivity, three-seed mean $\pm$ sd. Left: CAS macro-F1 and accuracy versus  $\pi_{\text{thr}}$  at  $B=15$ ; horizontal lines are the base weighted-TM. Right: versus the number of complementary pairs  $B$  at  $\pi_{\text{thr}}=0.8$ .

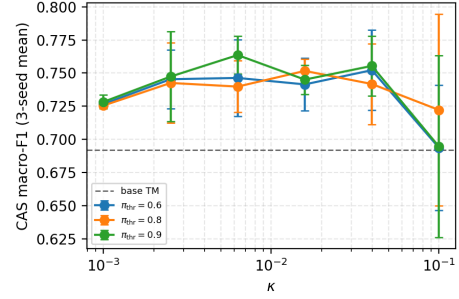


Fig. 4. CAS macro-F1 versus the CPSS inverse-regularization strength  $\kappa$  (log axis), one curve per  $\pi_{\text{thr}}$ , three-seed mean $\pm$ sd. The curve is flat over two decades and only collapses toward the base detector at  $\kappa=0.1$ .

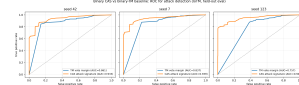


Fig. 5. ROC for attack detection on three seeded binary TMs. The CAS attack-signature score dominates the TM vote margin on every seed.

rebalances the operating point (recall 0.910, precision 0.947) and, more importantly, ranks flows far better, lifting AUROC from 0.819 to 0.940. (The binary arm re-binarizes at 5 bins versus 10; each arm is self-consistent.)

### E. Signature Content and Certificates

We now open the signatures up: how compact are they, what do they say, and how much can we trust? At  $\pi_{\text{thr}}=0.6$ , certified signature size  $|S|$  ranges from 115 to 311 clauses,  $|S^+|$  from 29 to 58, and the  $E[V]$  bound from  $\leq 72$  to  $\leq 250$ . At the verified  $\pi_{\text{thr}}=0.8$  operating point (Table IV), each family shrinks to just 28–53 sign-positive clauses while F1 improves — compaction and accuracy are not in tension here.

The content of the signatures is equally reassuring: the decoded DDoS signature (seed 42) reads like the classic flooding profile: 8 of its 29 sign-positive clauses are high-rate thresholds ( $\text{Rate} \geq 31.5\text{k packets/s}$ ), two more combine high rates with short headers, and two test for ICMP presence — matching the SYN-flag pattern found by the binary attack signature (§IV-D). Exculpatory lists, conversely, are dominated by long, low-rate conjunctions that look like benign device chatter — an analyst handed these lists would recognize the attack families from the rules alone.

TABLE II

$\kappa \times \pi_{\text{thr}}$ : THREE-SEED MEAN CAS MACRO-F1. THE BOTTOM ROW IS THE UNION-OVER-GRID RULE ACTUALLY USED, WHICH DOMINATES EVERY SINGLE  $\kappa$  AT THE VERIFIED  $\pi_{\text{thr}}=0.8$ .

| $\kappa \setminus \pi_{\text{thr}}$ | 0.5   | 0.6   | 0.7   | 0.8          | 0.9   |
|-------------------------------------|-------|-------|-------|--------------|-------|
| $1.0 \times 10^{-3}$                | 0.730 | 0.727 | 0.725 | 0.725        | 0.728 |
| $2.5 \times 10^{-3}$                | 0.748 | 0.745 | 0.740 | 0.743        | 0.747 |
| $6.3 \times 10^{-3}$                | 0.751 | 0.746 | 0.737 | 0.740        | 0.764 |
| $1.6 \times 10^{-2}$                | 0.748 | 0.741 | 0.751 | 0.752        | 0.745 |
| $4.0 \times 10^{-2}$                | 0.744 | 0.752 | 0.747 | 0.742        | 0.755 |
| $1.0 \times 10^{-1}$                | 0.674 | 0.694 | 0.718 | 0.722        | 0.695 |
| union grid (method)                 | 0.708 | 0.688 | 0.743 | <b>0.765</b> | 0.762 |

TABLE III

BINARY ATTACK DETECTION, THREE SEEDS (MEAN).

| Method               | attack F1                           | R / P         | AUROC        |
|----------------------|-------------------------------------|---------------|--------------|
| Binary TM argmax     | $0.912 \pm 0.000$                   | 0.988 / 0.847 | 0.819        |
| <b>CAS signature</b> | <b><math>0.928 \pm 0.007</math></b> | 0.910 / 0.947 | <b>0.940</b> |

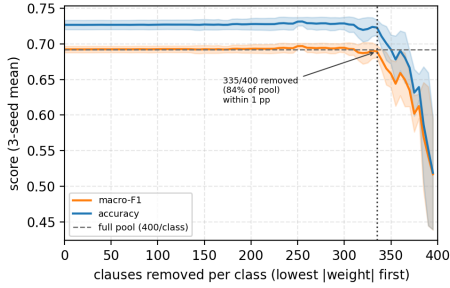


Fig. 6. Clause-weight pruning on the trained weighted TMs, three-seed mean $\pm$ sd on the full test set. Clauses are removed per class in order of increasing  $|\text{weight}|$ , five at a time. Macro-F1 and accuracy are flat until 84% of the pool has been deleted.

That the pool tolerates this much compaction is a property of the trained TM itself. Solving the pruning program of §II-E on these models, Fig. 6 traces the realised macro-F1 as the budget  $k$  grows in steps of five: it is flat until 335 of the 400 clauses per class (84% of the pool) are gone, so  $\varepsilon = 0.01$  gives  $k^* = 335$ .

The surrogate certifies but does not solve (1), and the slack is real: allocating the budget more cleverly does not help. Replacing the uniform per-class  $k$  by a global knapsack on  $\sum_c R(D_c)$ , or by the minimax rule equalising  $\max_c R(D_c)$ , changes macro-F1 by less than the cross-seed spread at matched pool size (0.689, 0.678 and 0.683 at 390 clauses).  $R$  is a worst case over inputs that never occur, so tightening its allocation does not track the realised objective.

#### F. CAS on a Weight-Pruned Pool

The pruning curve suggests an obvious question: if 84% of the pool is redundant for the base detector, is it also diluting the stability vote? We therefore keep only the top 65 clauses per class by  $|\text{weight}|$  — the knee of Fig. 6 — and re-run

TABLE IV

SIGNATURE COMPACTION AT THE VERIFIED OPERATING POINT (SEED 42,  $\pi_{\text{thr}}=0.8$ ). ALL FAMILIES START FROM THE SAME 2,400-CLAUSE POOL.

| Family   | Original clauses | Certified $ S $ | Final $ S^+ $ |
|----------|------------------|-----------------|---------------|
| Benign   | 2,400            | 184             | 41            |
| DDoS     | 2,400            | 63              | 29            |
| DoS      | 2,400            | 247             | 33            |
| MQTT     | 2,400            | 145             | 53            |
| Recon    | 2,400            | 66              | 28            |
| Spoofing | 2,400            | 194             | 50            |

TABLE V

CAS ON THE FULL VERSUS THE WEIGHT-PRUNED CLAUSE POOL, THREE-SEED MACRO-F1 AT  $B=15$ . PRUNING KEEPS THE TOP 65 CLAUSES PER CLASS.

|                                        | full pool (2,400)                   | pruned pool (390)                   |
|----------------------------------------|-------------------------------------|-------------------------------------|
| Weighted-TM argmax (base)              | $0.692 \pm 0.004$                   | $0.689 \pm 0.008$                   |
| CAS $\pi_{\text{thr}}=0.5$             | $0.708 \pm 0.011$                   | $0.777 \pm 0.023$                   |
| CAS $\pi_{\text{thr}}=0.6$             | $0.688 \pm 0.032$                   | <b><math>0.779 \pm 0.025</math></b> |
| CAS $\pi_{\text{thr}}=0.7$             | $0.743 \pm 0.015$                   | $0.778 \pm 0.023$                   |
| CAS $\pi_{\text{thr}}=0.8$             | <b><math>0.765 \pm 0.029</math></b> | $0.778 \pm 0.018$                   |
| CAS $\pi_{\text{thr}}=0.9$             | $0.762 \pm 0.038$                   | $0.779 \pm 0.016$                   |
| mean $ S^+ $ at $\pi_{\text{thr}}=0.8$ | 43                                  | 29                                  |

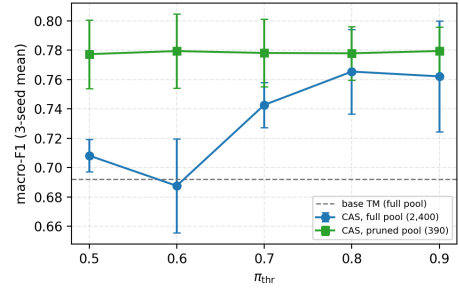


Fig. 7. CAS macro-F1 versus  $\pi_{\text{thr}}$  on the full and weight-pruned clause pools, three-seed mean $\pm$ sd. Pruning lifts the curve and flattens it: on the pruned pool the stability threshold no longer changes the result.

the identical CAS pipeline on the resulting 390-clause pool, changing nothing else.

The answer is yes, and by a clear margin (Table V, Fig. 7). Pruning costs the base detector nothing ( $0.689 \pm 0.008$  vs.  $0.692 \pm 0.004$ ), but CAS on the pruned pool reaches  $0.779 \pm 0.025$  — above the  $0.765 \pm 0.029$  obtained from the full pool — from signatures of only 26–38 clauses drawn from 390 rather than 2,400. More useful than the gain itself is what happens to the threshold: the full-pool sweep swings from 0.688 to 0.765 depending on  $\pi_{\text{thr}}$ , whereas the pruned pool is flat at 0.777–0.780 across the entire range, with roughly half the cross-seed spread at the top end. The  $\pi_{\text{thr}}$  instability documented in §IV-A is therefore a property of the diluted pool, not of CPSS: once the weak clauses are removed, the stability threshold stops mattering and the method no longer depends on tuning it.

The signatures compact in the same way (Table VI): cer-

TABLE VI

SIGNATURE COMPACTION ON THE WEIGHT-PRUNED POOL (SEED 42,  $\pi_{\text{thr}}=0.8$ ,  $B=15$ ). ALL FAMILIES START FROM THE SAME 390-CLAUSE POOL; COMPARE TABLE IV.

| Family   | Original clauses | Certified $ S $ | Final $ S^+ $ | $E[V]$    |
|----------|------------------|-----------------|---------------|-----------|
| Benign   | 390              | 63              | 30            | $\leq 32$ |
| DDoS     | 390              | 66              | 36            | $\leq 27$ |
| DoS      | 390              | 83              | 42            | $\leq 38$ |
| MQTT     | 390              | 64              | 26            | $\leq 30$ |
| Recon    | 390              | 58              | 22            | $\leq 41$ |
| Spoofing | 390              | 76              | 28            | $\leq 50$ |

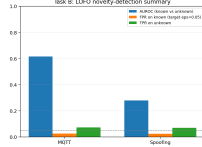


Fig. 8. Leave-one-family-out: conformal false-flag rates stay under target, but novelty AUROC against the held-out family is weak.

tified sets of 58–83 clauses and inculpatory cores of 22–42, comparable in absolute size to Table IV but drawn from a pool six times smaller, so far more of what survives pruning is actually used. The  $E[V]$  bounds tighten from  $\leq 72$ – $\leq 250$  to  $\leq 27$ – $\leq 50$ , the Shah–Samworth bound scaling with pool size  $p$ .

#### G. Task B: Open-Set Behavior (Leave-One-Family-Out)

A deployed detector will meet attacks it has never seen, so we stress-test CAS leave-one-family-out: hold out one family, retrain, and calibrate split-conformal novelty thresholds ( $\epsilon=0.05$ , two folds). The result is honestly negative: the false-flag guarantee holds in both folds (FPR 0.035 Spoofing, 0.027 MQTT, under target), but detection power against novel families is weak: Spoofing AUROC is 0.275, below chance, and MQTT reaches only 0.622 (Fig. 8). The reason is instructive: ARP-spoofing traffic closely resembles legitimate ARP-based device discovery in this feature space — a measured blind spot.

#### H. Signature Identifiability: Disjunct Certificates

The next question is combinatorial: if several attack families are active at once, can the signatures still be told apart? Arrange the six inculpatory signatures as columns of a binary clause-by-family table; the table is  $t$ -disjunct when no column is covered by the union of any  $t$  others, so every family keeps a unique mark under up to  $t$  simultaneous others. Computing the largest such  $t$  exactly gives  $\hat{t} = 5$  on every seed and both thresholds — the maximum possible with six families — so any mixture of up to five families decodes exactly when there is no noise.

Noise, however, separates the code from the decoder (500 trials per setting, three seeds): the strict COMP decoder recovers every mixture noiselessly, but collapses to 0.2%–1.8% under 5% bit-flip noise, while a thresholded variant

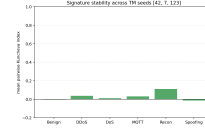


Fig. 9. Pairwise Kuncheva index on clause-index overlap between seeded TMs: near zero for every family.

that accepts partial matches restores 99.8%–100% decoding at threshold 0.8 (85%–90% at 0.9) and still reaches 68%–76% at 15% noise. Overlap reduction itself removes only 11–22 clauses from the pool. The code is strong; the decoder needs care under noise.

#### I. Cross-Seed Stability: A Negative Result

We close with a limitation: across the three seeded TMs, the pairwise Kuncheva index on clause-index overlap sits near zero for every family (−0.015 to 0.111; Fig. 9): identically numbered clauses in different seeds are unrelated learned rules, so CAS stabilizes selection *within* a trained TM; signature identity does not survive retraining.

### V. DISCUSSION

CAS improves on its base TM when the base still has headroom; a near-ceiling base would turn the same construction into lossy compression, so we position CAS as an attribution layer over a weak base detector. CICIoMT2024 models can suffer large cross-dataset F1 drops [13]; our claims remain single-dataset and transfer is open. Limitations: (i) no post-hoc XAI baseline (SHAP, Anchors) yet; (ii) loose  $E[V]$  bounds; (iii) instability at  $\pi_{\text{thr}}=0.6$  on the full pool, which weight-pruning removes (§IV-F); (iv) weak cross-seed signature identity (§IV-I); (v) synthetic OR-mixtures, not real traffic, in disjunct decoding.

### VI. CONCLUSION

CAS turns the clause pool of a trained TM into compact signed evidence lists per family, with a bound on spurious inclusions and exact mixture identifiability. On CICIoMT2024 it beats its base detector (macro-F1 0.765 vs. 0.692; binary AUROC 0.940); open-set and cross-seed limits bound the claim.

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